

RISK MANAGEMENT

STRATEGY:

Here we have used a hybrid strategy including both MACD and EMA with dynamic ATR band (Strategy 2 of last Assignment). For MACD, we have taken windows of (6,13,5) as it is very volatile at some places so that it can follow the quick changes...

The main idea is if macd line cross the signal line it will be buy and vice versa.

For EMA+ATR, We have taken window size of 7 (small), and specified the upper band and lower band ($EMA \pm (high - low)$)..

The main idea is that if price crosses above buy band it buy signal And vice versa for sell..

For the hybrid , It is simple that if both are satisfying buy and sell signal then only it will take a trade...

Then the trades are cleansed to keep one trade at a time ...

SIGNALS



RISK MEASURES AND BACKTESTING:

The main idea here is that if we are not in position later we can come across either buy or sell signal ,if it is buy/sell we will take a trade and set a stop loss of some % to that entry price....

But if we are in a position, let's say long, if the price goes below the stop loss created during entry or reaches a sell signal it will close the trade and will note down the tradewise metrics and vice versa for short trade...

Then to maintain the portfolio,in long, if it doesn't touch stop loss then it will update the capital and for short it will just add the profit or loss during the position as in short trade we do not invest our capital(in real)....

Metrics(with stop loss):

```
Returns: 522.1993146972657 %
Benchmark Return (in amount): 203550.5869140625

Number of closed trades: 47

Max holding time: 165
Avg Holding time: 14.595744680851064

Gross Profit: 522199.3146972656
Net Profit: 521259.3146972656

Max drawdown (in %): 56.45109196184394
Avg drawdown (in %): 27.11339193028979

Max dip (in %): -56.45109196184394
Avg dip (in %): -22.820189158129175

Sharpe Ratio: 2.630594314270871
```

Metrics(without stop loss):

```
Benchmark return = 199150.98828125
Return = 405.2067974057555 %

Number of closed trades = 31

Maximum holding time = 165
Mean holding time = 24

Gross Profit = 386888.50048828125
Net Profit = 386268.50048828125

Average drawdown for the trade = 31.918615501362506 %
Maximum drawdown for the trade = 63.10546912944487 %

Average dip for the trade = -27.214957033453192 %
Maximum dip for the trade = -63.10546912944487 %
Sharpe_like Ratio = 2.151864333944195
```

